Privacy-Preserving Environmental Disease Risk Forecasting Using Machine Learning and Output Perturbation

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***Abstract*—Public-health risk modeling can combine environmental, demographic, and surveillance indicators to support exploratory assessment of disease vulnerability, but such data can contain sensitive attributes and downstream outcome variables. This paper presents a research demonstration platform for three-class infection-risk classification and interactive privacy–utility analysis. The system uses a dataset described in the project materials as containing 43,689 records and 35 attributes, a leakage-aware feature policy, deterministic preprocessing, and a scikit-learn pipeline with a class-weighted Random Forest classifier. A React-based research interface communicates with a Python FastAPI backend to support dataset exploration, visualization, model training, individual risk simulation, and privacy-parameter experiments. The project-reported baseline test accuracy is 54.57%, while an output-perturbed configuration with $\epsilon = 1$ and $\delta = 10^{-5}$ reports 31.92% accuracy. The decrease illustrates the practical utility cost of strong output randomization in this implementation. However, because the supplied implementation description does not establish a verified sensitivity bound for the released probability vector or a formal end-to-end privacy analysis of model training and repeated queries, the mechanism is treated as a privacy-oriented prototype rather than a proven differentially private learning system. The paper documents the architecture, methodology, reported results, limitations, and steps required for a rigorous future evaluation.**

***Index Terms*—Differential privacy, disease-risk classification, environmental health, Random Forest, privacy–utility trade-off, FastAPI, public-health analytics**

I. Introduction

Public-health surveillance increasingly uses statistical and machine-learning methods to identify patterns in health outcomes and environmental conditions. Temperature, humidity, air quality, population density, and surveillance counts can be relevant contextual indicators for understanding disease burden. At the same time, health-related datasets may contain attributes that identify or characterize individuals, and model outputs can create privacy risks if they reveal information about records used during development. Differential privacy (DP) provides a mathematical framework for bounding the effect of an individual record on the output of a randomized computation [1]. Its application to machine learning requires careful attention to the algorithm, sensitivity assumptions, privacy accounting, and the complete set of released outputs [2].

This work presents a full-stack research demonstration named *Privacy-Preserving Environmental Disease Risk Forecasting*. It is intended as an interactive workbench rather than a deployed clinical decision-support system. The platform combines dataset inspection and sanitization, a supervised three-class classifier, environmental visualizations, and an interface for exploring output perturbation. The project materials report a dataset of 43,689 rows and 35 columns, with the target variable `Infection_Risk_Level` divided into Low, Medium, and High Risk.

The contributions of this paper are fourfold: (1) a documented data-preparation and feature-governance pipeline that excludes specified sensitive and leakage-prone fields; (2) a reproducible software architecture that separates a React frontend from a FastAPI backend; (3) a baseline Random Forest evaluation and a reported comparison with noisy probability outputs; and (4) a critical discussion of why the implemented perturbation should not be interpreted as a formal end-to-end DP guarantee without further analysis.

The remainder of this paper reviews related work, describes the dataset and system, details the modeling and privacy methodology, reports the available results, and discusses limitations and future work.

II. Related Work

A. Differential Privacy

Dwork and Roth formalize DP as a guarantee that the output distribution of a randomized mechanism changes only by a bounded amount when one record is added or removed from a dataset [1]. The parameters ϵ and δ govern the strength of the guarantee: smaller ϵ generally provides stronger protection, while δ allows a small probability of failure. A Gaussian mechanism can provide approximate DP when its noise scale is calibrated to a valid sensitivity bound and the relevant conditions are satisfied. The privacy guarantee follows from this calibration, not simply from adding Gaussian noise.

Abadi *et al.* demonstrated a practical route to differentially private model training using gradient clipping, noise addition, and privacy accounting [2]. That approach differs from adding noise to predictions of a model trained conventionally: output perturbation protects a particular release only when the released function's sensitivity is bounded with respect to the neighboring datasets under consideration. A privacy claim therefore needs to specify the protected unit, adjacency relation, mechanism, sensitivity, and query or release budget.

B. Machine Learning for Health Data

Privacy-preserving methods have been studied for medical data analysis, where predictive utility must be balanced against

disclosure risk [3]. The literature emphasizes that privacy mechanisms can reduce model utility and that the magnitude of the reduction depends on the data, task, privacy budget, and mechanism. This motivates reporting more than a single accuracy value, including class-wise recall and macro-averaged metrics where class frequencies differ.

C. Random Forest Classification

Random Forests combine predictions from multiple decision trees trained with randomized sampling and feature selection [4]. They are practical for mixed tabular features and can capture nonlinear interactions without requiring feature scaling. In this project, a class-weighted Random Forest provides a baseline classifier. Class weighting can influence training behavior under imbalance, but it does not itself guarantee balanced class-wise performance; evaluation must inspect each class separately.

D. Model Disclosure and Federated Learning

The privacy threat model is informed by work on membership inference and training-data extraction. Shokri *et al.* showed that model responses can leak whether an example was part of training [5]. Carlini *et al.* demonstrated extraction of memorized training data from large language models [6]. Although the latter setting differs from tabular environmental classification, both lines of work motivate treating model outputs as a possible disclosure channel rather than assuming that data are safe once direct identifiers have been removed. These papers motivate future adversarial testing; they do not establish that an attack has succeeded against the present prototype.

Federated learning offers another design direction when data are distributed across organizations. McMahan *et al.* introduced communication-efficient training from decentralized data [7]. Keeping raw records local can reduce central data movement, but model updates may still reveal information and therefore do not replace a formal privacy mechanism. For this project, federated learning is a possible extension rather than an implemented capability. The current prototype instead uses a centralized dataset and explores output perturbation, so the two approaches should not be conflated.

E. Mechanism Calibration and Practical Evaluation

Balle and Wang showed that Gaussian noise calibration can be improved through analytical methods [8], while Ji and Li examined alternatives for high-dimensional output perturbation and its utility costs [9]. These studies reinforce that a noise formula is meaningful only when paired with a clearly defined query and a defensible sensitivity bound. For probability vectors, clipping and renormalization are useful to produce valid probabilities, but post-processing cannot correct an invalid privacy calibration. Jayaraman and Evans evaluated the practical utility of differentially private machine learning and emphasized the need to report both privacy and predictive performance [10]. Accordingly, the present work reports class-sensitive metrics and treats its observed utility reduction as a preliminary experiment rather than a universal conclusion about differential privacy.

III. System Overview

A. Architecture

The application uses a decoupled client–server design. The frontend is implemented with React 19 and Vite and presents seven views: Dashboard, Dataset Explorer, Visualizations, Model Training, Risk Prediction, Privacy Preserving, and Results. The backend is implemented in Python with FastAPI and exposes REST endpoints for dataset summaries, preview and filtering, preprocessing, chart aggregation, model training, prediction, and privacy experiments. Axios connects the frontend to the backend, while Recharts renders interactive plots. Model and result artifacts are stored using Joblib and JSON, according to the supplied project description.

The data flow begins with the original CSV file. The preprocessing module creates a cleaned and sanitized dataset. The model module fits a preprocessing-and-classification pipeline and stores the resulting artifacts. The privacy module evaluates perturbed probability vectors, and the visualization module aggregates data for charts. The frontend provides a common interface to these backend functions. This architecture is useful for demonstrating an end-to-end research workflow, but the application itself does not establish the clinical validity or deployment readiness of its predictions.

B. Dataset and Target Distribution

The project description identifies the input file as `public_health_surveillance_dataset.csv`. It reports 43,689 records and 35 attributes. The provenance, collection protocol, geographic coverage, time span, and licensing details were not included in the materials available for this paper; these must be documented before external publication or reuse.

The target counts supplied by the project are shown in Table I. The class distribution is imbalanced, with Low Risk accounting for approximately 60.30% of the records and High Risk for approximately 10.14%. These proportions make accuracy alone insufficient to assess minority-class performance.

TABLE I
Project-Reported Target Distribution

Risk class	Records	Share
Low Risk	26,346	60.30%
Medium Risk	12,912	29.55%
High Risk	4,431	10.14%
Total	43,689	99.99%[a]

[a] The total is 99.99% because the displayed class percentages are rounded. Counts and percentages are reproduced from the supplied project description.

C. Feature Governance

The project configuration defines 25 approved predictor features: 12 numerical and 13 categorical. Numerical variables include age, chronic-condition count, vaccination status,

guideline compliance, temperature, air-quality index (AQI), humidity, transmission rate, mortality rate, resource utilization, daily new cases, and case-fatality ratio. Categorical variables include gender, location, socioeconomic status, immunity level, reported symptoms, testing results, population density, travel history, social activity, vaccination hesitancy, hospitalization rate, hospital capacity, and healthcare-personnel availability.

The project excludes `Ethnicity`, `Medical_History`, and `Diagnosis` from the model feature set and masks them by default in summaries. It also excludes `Disease_Severity`, `Hospitalization_Requirement`, and `Outbreak_Status` to reduce the risk of target leakage from downstream outcomes. Date fields are parsed and are not used as ordinary static categorical predictors. These are project-specific governance choices; they should be revisited with domain experts to ensure that the remaining predictors are available at the intended prediction time and do not encode post-outcome information indirectly.

IV. Data Preprocessing

The preprocessing module is described as performing the following steps. First, exact duplicate rows are removed; the supplied review reports zero duplicates in the original dataset. Date columns are parsed to datetime values, with invalid values coerced to missing values. Medical history, reported symptoms, and diagnosis fields are described as having 60%, 50%, and 70% missingness, respectively, and are imputed with constant values such as "Unknown" or "None" for data exploration. Remaining numerical fields are imputed using the median, and remaining categorical fields use "Unknown" or the most frequent value depending on the pipeline stage.

Range validation is applied to environmental fields. Temperature values outside $[-50, 60]$ degrees Celsius are replaced with the median; AQI values are checked against $[0, 500]$; and humidity is checked against $[0, 100]\%$. Leading and trailing whitespace is stripped from categorical strings. The pipeline exports a cleaned CSV for downstream analysis.

For model fitting, the project description states that imputation and encoding are encapsulated in a scikit-learn pipeline with a `ColumnTransformer`. Numerical fields use median imputation. Categorical fields use most-frequent imputation followed by an ordinal encoder configured to map unseen categories to -1. Keeping learned preprocessing steps inside the training pipeline reduces the risk of information leakage from the test set. For a fully reproducible evaluation, imputation statistics and encoders should be fitted only on training data and then applied unchanged to the held-out set.

V. Modeling Methodology

A. Random Forest Pipeline

The classifier is a `RandomForestClassifier` with 150 trees, `min_samples_split`=5, `min_samples_leaf`=2, `class_weight`=`balanced`, `random_state`=42, and parallel execution. The target is encoded as Low Risk (0), Medium Risk (1), and High Risk (2). A stratified 80:20 train–test split is reported, producing 34,951 training samples and 8,738 test samples. The classifier is intended to provide a baseline for three-class risk categorization, not a causal estimate or validated medical diagnosis.

B. Evaluation Metrics

The system reports accuracy, macro precision, macro recall, macro F1-score, class-specific recall, and a confusion matrix. Accuracy is the fraction of correctly classified records. Macro precision, recall, and F1 average the metric calculated separately for each class, giving each class equal weight regardless of prevalence. This is important because the target classes are imbalanced. A complete evaluation should additionally report confidence intervals, repeated or temporal validation, calibration, and subgroup performance; those analyses were not included in the supplied results.

VI. Privacy Mechanism

A. Differential Privacy Definition

For neighboring datasets D and D' that differ by one protected record, a randomized mechanism $\mathcal{M}$ satisfies (ϵ, δ)-DP if, for every measurable output set S,

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\epsilon} \Pr[\mathcal{M}(D') \in S] + \delta. \tag{1}$$

This definition describes a guarantee over output distributions. It does not follow merely from hiding selected columns in a user interface, and it does not automatically protect an unmodified model trained on sensitive records.

B. Implemented Output Perturbation

The supplied project description reports adding Gaussian noise to a model's class-probability vector $\mathbf{p} = [p_0, p_1, p_2]$:

$$\widetilde{\mathbf{p}} = \mathbf{p} + \mathcal{N}(0, \sigma^2 I), \tag{2}$$

where the described calibration is

$$\sigma = \Delta f \frac{\sqrt{2 \ln(1.25/\delta)}}{\epsilon}. \tag{3}$$

The project uses a conservative configured value $\Delta f = 1.0$ and $\delta = 10^{-5}$. At $\epsilon = 1$, this expression gives $\sigma \approx 4.845$. Negative values are clipped to zero and the vector is normalized to sum to one:

$$\widehat{p}_i = \frac{\max(0, \widetilde{p}_i)}{\sum_j \max(0, \widetilde{p}_j)}. \tag{4}$$

This describes the implemented experiment, but two caveats are essential. First, the materials provided do not establish that the class-probability function has the assumed global sensitivity $\Delta f = 1$ with respect to a one-record change in the training data. Second, the baseline model is described as conventionally trained, with noise added to its probability output. Without a proof of sensitivity and a clear accounting of all releases and repeated queries, the implementation cannot be claimed to provide a formal end-to-end DP guarantee for the training dataset. Projection and normalization are post-processing operations, which preserve an already-valid DP

guarantee, but cannot create one when the preceding mechanism has not been shown to satisfy it. We therefore refer to this component as an output-perturbation prototype and treat its privacy–utility results as empirical, not as a certified privacy guarantee.

VII. Experimental Results

A. Reported Baseline and Perturbed Results

The following values are reproduced from the project review supplied for this paper. The review reports a baseline accuracy of 55.37% in one summary and a test accuracy of 54.57% in the evaluation description. Table II uses the explicit test accuracy for the baseline comparison and retains the reported privacy-perturbed accuracy. The discrepancy between baseline summary and test values should be resolved by regenerating the metrics from the saved results file before publication.

TABLE II
Project-Reported Baseline and Output-Perturbation Metrics

Metric	Baseline	Perturbed
Accuracy	54.57% (test)	31.92%
Macro precision	0.3314	0.3348
Macro recall	0.3337	0.3353
Macro F1-score	0.3049	0.2920
Low-risk recall	84.06%	31.77%
High-risk recall	0.34%	39.05%

The reported perturbed accuracy is 22.65 percentage points below the stated baseline test accuracy. Macro precision increases slightly by 0.0034 and macro recall by 0.0016, while macro F1 decreases by 0.0129. The reported high-risk recall rises from 0.34% to 39.05%, while low-risk recall falls from 84.06% to 31.77%. The increase in high-risk recall should not be interpreted as an unqualified improvement: strong randomization can distribute predictions across classes and may increase a minority-class recall while substantially reducing overall discrimination. Confusion matrices, per-class precision, repeated random seeds, and calibration analyses are needed to determine whether the behavior is robust.

The reported noise standard deviation of approximately 4.845 is large relative to probabilities on the interval $[0, 1]$. Consequently, clipping and renormalization can produce distributions that differ substantially from the original model outputs. This provides a plausible explanation for the sharp utility reduction, but the exact effect depends on the random noise draws and implementation details. The results should be reproduced over multiple runs and reported with variability before drawing general conclusions.

B. Interpretation

The experiment illustrates that privacy-oriented randomization can materially alter a model's predictive behavior. It also demonstrates why aggregate accuracy should not be used alone for imbalanced risk categories. At the same time, the current results are insufficient to establish that the reported metric changes are caused solely by a specific formal privacy level: the guarantee itself has not been verified, and the materials do not provide raw prediction outputs, the complete confusion matrices, or uncertainty estimates. The evidence therefore supports a preliminary software demonstration and a qualitative privacy–utility observation, not a claim of clinically useful forecasting or proven DP learning.

VIII. Research Interface

The dashboard presents total records, missingness summaries, and target distribution. The Dataset Explorer supports pagination, searching, sorting, data-quality inspection, and default masking of selected sensitive fields. The visualization view provides environmental distributions and relationships, temporal and regional summaries, socioeconomic breakdowns, and a correlation matrix. The Model Training view triggers model fitting and displays evaluation summaries and feature importances. The Risk Prediction view accepts selected environmental and demographic values and returns a predicted class and class probabilities. The Privacy Preserving view exposes ϵ and δ controls and compares reported metrics. The Results view collects model metrics and class-wise evaluation summaries.

These functions make the project suitable for demonstration and experimentation. Since the application presents predictions in a health-related context, its interface should consistently label outputs as research simulations, communicate uncertainty, avoid implying individual diagnosis, and avoid deployment for operational public-health decisions until the data and model are independently validated.

A. Operational Design and Responsible Use

The React 19 and Vite frontend is separated from the Python FastAPI service. Axios provides the client-side API layer, while the backend separates configuration, dataset profiling, preprocessing, model training, privacy logic, visualization aggregation, and endpoint routing. This modular design improves maintainability because data preparation, modeling, and privacy-related behavior can be reviewed independently. The frontend provides seven views: Dashboard, Dataset Explorer, Visualizations, Model Training, Risk Prediction, Privacy Preserving, and Results.

For a research prototype, the interface should make uncertainty and limitations visible at the point where a result is displayed. In particular, a predicted category should not be presented as a diagnosis or an individualized medical recommendation. The system should indicate that inputs are simulated, show class probabilities without overstating calibration, and distinguish descriptive associations from causal conclusions. Any future deployment would require review by epidemiologists, privacy specialists, security engineers, and appropriate institutional governance bodies.

Data minimization should also be enforced at the API boundary rather than only in the visual interface. Sensitive columns are described as masked by default and excluded from the model feature set, but masking in a table is not equivalent to access control if the underlying endpoint still returns the

values. The production-oriented version should verify response schemas, remove disallowed fields before serialization, enforce authorization, apply transport security, log access without recording sensitive payloads, and define retention policies. Public CSV exports should be reviewed separately from the on-screen preview because exports can bypass visual masking.

IX. Extended Analysis and Validation Plan

A. Feature Governance and Temporal Availability

Feature governance is central to the validity of risk classification. The project defines 25 approved predictors, comprising 12 numerical and 13 categorical variables. Numerical inputs include age, chronic conditions, vaccination status, guideline compliance, temperature, air-quality index, humidity, transmission rate, mortality rate, resource utilization, daily new cases, and case-fatality ratio. Categorical inputs include gender, location, socioeconomic status, immunity level, reported symptoms, testing results, population density, travel history, social activity, vaccination hesitancy, hospitalization rate, hospital capacity, and healthcare-personnel availability. The feature list is a software configuration decision; it should not be interpreted as proof that every variable is appropriate for every prediction horizon.

A forecasting system must define the moment at which a prediction is made and the information available at that moment. For example, daily new cases may be available for a regional forecast after reporting delays, while a case-fatality ratio may be computed from outcomes that are not yet known at the beginning of the forecast period. If the model uses information recorded after the target event, the evaluation can be optimistic even when explicitly named leakage columns are excluded. A future version should therefore document a prediction timestamp, an outcome window, feature provenance, and the earliest valid availability time for each predictor. Features should be removed or lagged when their timing is incompatible with the intended forecast.

The distinction between prediction and explanation also matters. Random Forest feature importance describes how the fitted model uses the supplied variables under a particular training procedure; it does not establish that changing a feature would change disease risk. Correlated predictors can share or distort importance, and a high importance score may reflect reporting practices, geographic differences, or artifacts in the target-generation process. Feature importance should consequently be presented as exploratory model diagnostics, not causal evidence or policy guidance.

B. Data Quality and Preprocessing Decisions

The preprocessing pipeline removes exact duplicate rows, parses date fields, imputes missing values, validates environmental ranges, and trims categorical whitespace. The project description reports no exact duplicate rows in the initial dataset. Medical history, symptoms, and diagnosis have substantial missingness, reported as approximately 60%, 50%, and 70%, respectively. Those columns are excluded from the approved feature set, but missingness still matters for dataset summaries and any public exports. Missingness may be systematic rather than random: it can reflect access to care, local reporting capacity, or a data-collection workflow. A future analysis should compare missingness across target classes, geography, and time, while ensuring that such comparisons do not disclose small groups.

Median imputation for numerical variables is robust to some extreme values, while most-frequent or constant imputation provides a simple treatment for categorical values. These choices are practical baselines, not universally optimal methods. All imputation statistics and encoders must be fitted using only the training partition and then applied unchanged to the test partition. Encapsulating preprocessing in a scikit-learn pipeline helps enforce this boundary and reduces the risk that information from the test set influences training. The project should retain this pipeline-based design as the preferred approach rather than preprocessing the full dataset before splitting.

The range checks for temperature, humidity, and air-quality index introduce domain constraints. Such checks should be accompanied by a record of how many values are modified and why, since silently replacing invalid values can hide data ingestion defects. It is preferable to preserve an audit summary containing counts before and after cleaning, rules applied, and a version identifier for the cleaned output. The interface already aims to display before-and-after quality statistics; those summaries should be generated from actual processing results rather than hard-coded sample values.

C. Metrics for Imbalanced Classification

The target distribution is imbalanced: 26,346 records are labeled Low Risk, 12,912 Medium Risk, and 4,431 High Risk. A classifier can achieve a seemingly reasonable overall score by favoring the majority class while failing to identify the highest-risk cases. This is why the evaluation includes macro precision, macro recall, macro F1-score, and per-class recall alongside accuracy. Macro averaging assigns equal weight to each class and can expose weaknesses that a prevalence-weighted metric may hide.

The project-reported baseline test accuracy is 54.57%, while a separate project summary reports 55.37%. This inconsistency must be resolved from the saved metric artifact before a submission or comparison claims a single baseline value. The table in this paper uses the test value reported in the detailed results description and explicitly preserves the discrepancy as a limitation. It does not infer an unreported confusion matrix or invent precision and recall values for individual classes.

For a more rigorous study, the evaluation should include a normalized confusion matrix, class-specific precision and recall, macro and weighted F1, balanced accuracy, and probability calibration. Calibration can be assessed with reliability diagrams and a proper scoring rule, provided that the test set is sufficiently representative. Confidence intervals can be estimated through bootstrap resampling of held-out observations, while repeated training seeds can characterize model instability. If the data have a temporal structure, a chronological test

period is especially important because random splitting can allow neighboring time periods to appear in both training and testing partitions.

D. Comparative Privacy Experiments

A useful privacy–utility experiment should vary the privacy parameter across a pre-specified grid, for example several values of ϵ while keeping δ and the data split fixed. For each setting, the study should record the noise scale, accuracy, macro F1, class-wise recall, calibration, and run-to-run variation. Results should be plotted as a trade-off curve rather than presenting only one configuration. Lower ϵ generally corresponds to a stronger privacy constraint and often requires more noise, but the utility impact depends on the mechanism, sensitivity, data distribution, and task.

The present output perturbation experiment sets $\Delta f = 1.0$, $\epsilon = 1.0$, and $\delta = 10^{-5}$. Using the classical Gaussian expression stated in the project materials gives a noise standard deviation of approximately 4.845. Since each unperturbed class probability is between zero and one, this scale can dominate the original signal. The resulting probability vector is clipped at zero and normalized to sum to one. The reported accuracy falls to 31.92%, indicating a substantial loss of predictive utility in this configuration. The comparison is informative as an implementation demonstration, but it is not sufficient to prove that the privacy guarantee is valid or that this level of noise is necessary for a correctly calibrated mechanism.

The normalization step is post-processing of the noisy vector. Under the post-processing property, it does not weaken a valid DP guarantee. However, this property applies only if the pre-normalized output was already generated by a mechanism satisfying the required guarantee. The current study does not provide a proof that the classifier's probability vector has global sensitivity at most one with respect to the stated neighboring-dataset relation. Thus, the reported ϵ and δ values should be treated as configuration inputs, not as a certified end-to-end privacy statement.

E. Research Ethics, Governance, and Deployment Boundaries

Health-related prediction tools should be governed according to the potential consequences of errors. A false low-risk prediction may delay attention to a vulnerable population, while a false high-risk prediction can divert scarce resources or create unnecessary concern. A responsible evaluation should identify the intended decision-maker, the action that might follow a prediction, the costs of different errors, and the process for human review. In this project, the model is a demonstration and should not be used to diagnose individuals or direct clinical care.

Privacy engineering should be combined with access control, data minimization, secure storage, transport encryption, retention limits, and documented incident response. A privacy mechanism does not protect against every risk: it does not ensure that the source data were lawfully collected, prevent unauthorized access to raw files, eliminate bias, or guarantee the validity of a prediction. The application should separate public demonstration data from sensitive operational data and should not expose raw patient-level records through a public API. If real personal health information is ever used, institutional review and applicable legal and ethical requirements must be addressed before data collection or deployment.

The proposed future evaluation should be staged. First, verify the dataset's provenance, license, label construction, and data dictionary. Second, regenerate baseline metrics and resolve the accuracy discrepancy. Third, perform temporal and subgroup validation, including uncertainty estimates. Fourth, define patient-level adjacency and formally calibrate or replace the privacy mechanism. Fifth, conduct membership-inference and other appropriate privacy tests as empirical diagnostics, while recognizing that attack failure does not substitute for a formal proof. Finally, obtain independent review of security, fairness, usability, and intended-use limitations before considering any operational application.

X. Limitations and Threats to Validity

Several limitations constrain interpretation. First, the dataset's provenance and representativeness are not established in the supplied description. Without documentation of collection, labels, geography, time period, sampling, and licensing, external validity cannot be assessed. Second, the reported baseline accuracy differs between the project summary and test evaluation (55.37% versus 54.57%). Third, the current evaluation lacks repeated runs, confidence intervals, a complete confusion matrix, calibration analysis, and temporal or external validation. Fourth, the listed features include surveillance-derived quantities such as mortality rate, transmission rate, and case-fatality ratio; their availability at prediction time must be verified to prevent temporal or conceptual leakage. Excluding three named leakage fields does not by itself prove that the remaining features are leakage-free.

Fifth, output perturbation is not yet supported by a formal sensitivity proof or end-to-end privacy accounting. A rigorous design should define record-level or patient-level adjacency, bound the contribution of each protected unit, calibrate noise to a proven sensitivity, account for repeated predictions, and test attacks or use a mechanism with an established guarantee. If model training itself is to be protected, differentially private training methods such as DP-SGD or an appropriately analyzed private learning algorithm should be considered [2], [7], [8], [10]. Sixth, the system is an academic prototype and has not been reported as having undergone clinical validation, security testing, fairness auditing, or deployment evaluation.

XI. Conclusion and Future Work

This paper documented a full-stack platform for environmental disease-risk classification and privacy–utility exploration. The system combines data sanitization, feature governance, a class-weighted Random Forest pipeline, interactive visualization, and a Gaussian output-perturbation experiment. Project-reported results show a reduction in test accuracy from 54.57% to 31.92% under the configured noisy-output

experiment, alongside substantial changes in class-wise recall. These observations demonstrate the importance of measuring both utility and class-specific behavior when adding noise to risk predictions.

The next steps are to verify dataset provenance and licensing, regenerate all metrics from versioned artifacts, correct the baseline metric discrepancy, conduct repeated experiments, and document the full feature availability timeline. Most importantly, the privacy component should be upgraded or formally analyzed so that sensitivity, adjacency, privacy accounting, and repeated-query behavior support any stated DP guarantee. Future evaluation should include external or temporal validation, calibration, subgroup analysis, and a clear separation between research simulation and clinical use. Until these steps are completed, the platform should be described as a research prototype rather than a validated or formally certified privacy-preserving forecasting system.

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